

Janarthanan S

Vellore, Tamil Nadu

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Summary

B.E. graduate (May 2026) and Python-focused Software Engineer with hands-on experience building REST APIs with FastAPI and developing NLP/ML applications. Strong foundation in Python, backend development, SQL, machine learning, and DSA. Solved **600+ DSA problems** on LeetCode. Seeking a Software Engineer role to build reliable backend systems, APIs, and AI-driven applications.

Technical Skills

- **Languages:** Python, Java (Basic), SQL, HTML, CSS
- **AI / ML:** Machine Learning, NLP, TF-IDF, Text Preprocessing, Deep Learning, Pandas, NumPy, OpenCV, YOLOv11
- **Backend & APIs:** FastAPI, Flask, REST APIs, CRUD, Pydantic, SQLAlchemy ORM, Groq API
- **Databases:** PostgreSQL, SQLite, Firebase Firestore, Database Indexing, Query Optimization
- **Authentication:** JWT, OAuth2, bcrypt, Firebase Authentication
- **Cloud & DevOps:** AWS (EC2, S3, RDS, IAM), Docker, Docker Compose, Git, GitHub Actions
- **Testing & Tools:** Pytest, Postman, Swagger UI, pgAdmin, VS Code
- **CS Concepts:** OOP, DSA, OS, Computer Networks, DBMS, Basic System Design

Internship Experience

Proxenix Software Company

July 2025 – September 2025

Software Engineer Intern — Backend & ML (Remote)

- Developed a sentiment analysis system using Python, TF-IDF, and text preprocessing to classify customer and movie reviews as positive or negative.
- Built FastAPI REST endpoints and integrated the trained machine learning model to process client requests and return real-time sentiment predictions.
- Implemented an end-to-end ML workflow covering data preprocessing, feature extraction, model training, evaluation, and prediction while collaborating with the development team.

Projects

Smart Hostel Issue Tracking System | Python, Flask, Flask-SQLAlchemy, SQLite, HTML, CSS | GitHub | Live Demo Feb 2026

- Developed a web-based hostel issue tracking system to digitize complaint submission, tracking, and resolution for students and hostel management.
- Designed relational data models for users and complaints using Flask-SQLAlchemy and SQLite, supporting structured issue management and user-specific records.
- Implemented Flask routes and role-based workflows for creating, viewing, updating, and tracking complaint status throughout the resolution process.

Pothole Detection & Real-Time Alert System | Python, YOLOv11, OpenCV, Deep Learning | GitHub

Nov 2025

- Developed a real-time pothole detection system using YOLOv11 and OpenCV to identify road hazards from images and video streams for automated road-safety monitoring.
- Prepared and augmented the pothole dataset and tuned model training parameters, achieving **92% precision** on evaluation data while reducing false-positive detections.
- Integrated the trained YOLOv11 model with OpenCV-based image and video processing to perform frame-by-frame inference and generate pothole detection results.

AIjust Spoken English Platform | Firebase, Firestore, Groq API, LLM, Speech Recognition, Text-to-Speech | GitHub

Jan 2026

- Developed an AI-powered spoken English platform combining speech recognition, text-to-speech, and LLM-based conversational feedback for interactive language practice.
- Integrated the Groq API with Llama 3.3 70B to generate conversational responses and grammar corrections from user speech transcripts.
- Implemented Firebase Authentication and Firestore for user accounts and persistent correction history, supporting email/password and Google Sign-In.

Education

Madras Institute of Technology, Anna University

Sep 2023 – May 2026

B.E. Electronics and Communication Engineering – CGPA: 7.28/10

Chennai, Tamil Nadu

Rajagopal Polytechnic College

2020 – 2023

Diploma in Electronics and Communication Engineering – 94.30%

Vellore, Tamil Nadu

Achievements & Certifications

- Solved **600+ DSA problems** on LeetCode using Python, demonstrating strong algorithmic problem-solving skills.
- Grew **LinkedIn community to 2,000+ followers** sharing Python, backend, and software development content.
- **HackerRank:** Python (Basic) | SQL (Basic) | Problem Solving (Basic) | REST API (Intermediate)
- **Python for Beginners** – Scaler Academy | **Basics of Python** – Udemy

Leadership & Collaboration

- Led teams of 4–6 members through task allocation, sprint planning, and project delivery.
- Mentored peers in Python, DSA, and problem-solving while collaborating on technical issues.